



IoT & ML-based Vehicle Speed Monitoring with RFID-Enabled License Validation

PRESENTED BY
25-26J-314

Agenda

- 1 Introduction
- 2 Social & Research Impact
- 3 Previous Research & Shortcomings
- 4 Constraints & Limitations
- 5 Individual Component

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The Real-World Problem

- ❖ Road accidents in Sri Lanka: Over-speeding contributes to nearly 30% of deaths annually.
- ❖ Current Enforcement: Speed guns & CCTV = reactive, manual, limited coverage.
- ❖ Consequences:
 - Drivers → repeat offences due to lack of feedback.
 - Authorities → lack of real-time enforcement tools.

Why Solving This Problem Matters

Social Impact

- Safer roads → fewer injuries & deaths.
- Disciplined drivers through instant feedback.
- Reduced burden on traffic police.

Research Impact

- Combines IoT + ML + Visualization → contributes to smart traffic management research.
- Provides predictive analytics → supports policymaking & enforcement strategies.

Gaps in Existing Approaches

Previous Methods:

- RFID used for tolling in India, Singapore.
- CCTV-based monitoring with manual review.

Shortcomings:

- No prediction of repeat offenders / high-risk times.
- No integrated dashboard for visualization.
- No real-time alerts to drivers.

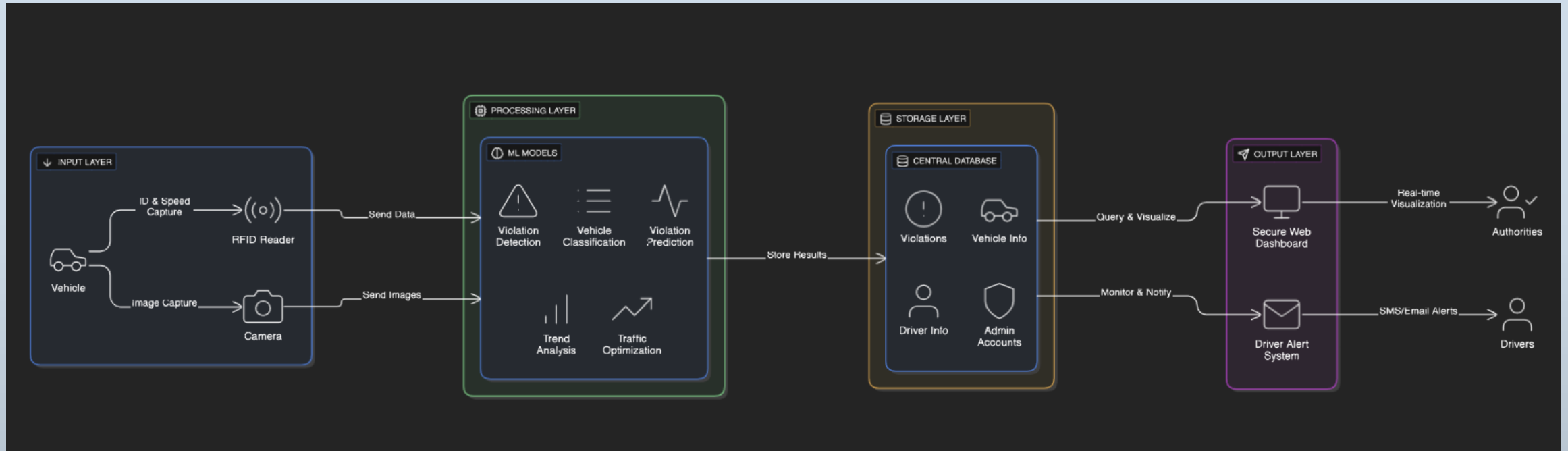
Our Approach:

- Integrated system to IoT sensors + ML models + Real-time Dashboard + Driver Alerts.

Constraints & Limitations

- Data Collection: Limited real-world violation datasets.
- Hardware Costs: RFID & IoT devices are expensive.
- Privacy Concerns: Driver data must remain secure & anonymized.
- Timeframe: Limited project duration restricts full deployment.

System Overview



Dynamic Roadside Behavior Analysis Model for Real-Time Traffic Violation Detection



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Dynamic Roadside Behavior Analysis Model for Real-Time Traffic Violation Detection

Identified Challenges

- Current speed enforcement is **reactive** (speed guns, cameras).
- Coverage is **limited**, monitoring is **manual**, no predictive insights.

Proposed Innovative Solution

- Apply **RFID-based speed tracking** combined with **ML models**.
- Predict **when, where, and who** is most likely to violate traffic rules.
- Enables **data-driven enforcement** and **proactive road safety**.

Limitations of Current Approaches

- Cannot identify **repeat offenders**.
- No prediction of **high-risk timeframes, vehicle types, or locations**.

Dynamic Roadside Behavior Analysis Model for Real-Time Traffic Violation Detection

Core Knowledge Areas

- Data Science & Machine Learning → vehicle classification, violation forecasting.
- Predictive Analytics → high-risk vehicles, times, and locations.
- Image Processing → vehicle type detection for accuracy.

Technologies Used

- RFID chips & high-sensitivity scanners → reliable data capture.
- ML Models: Random Forest (classification), Time-Series Forecasting (prediction).
- Visualization Tools → charts/graphs for predictive insights.

Dynamic Roadside Behavior Analysis Model for Real-Time Traffic Violation Detection

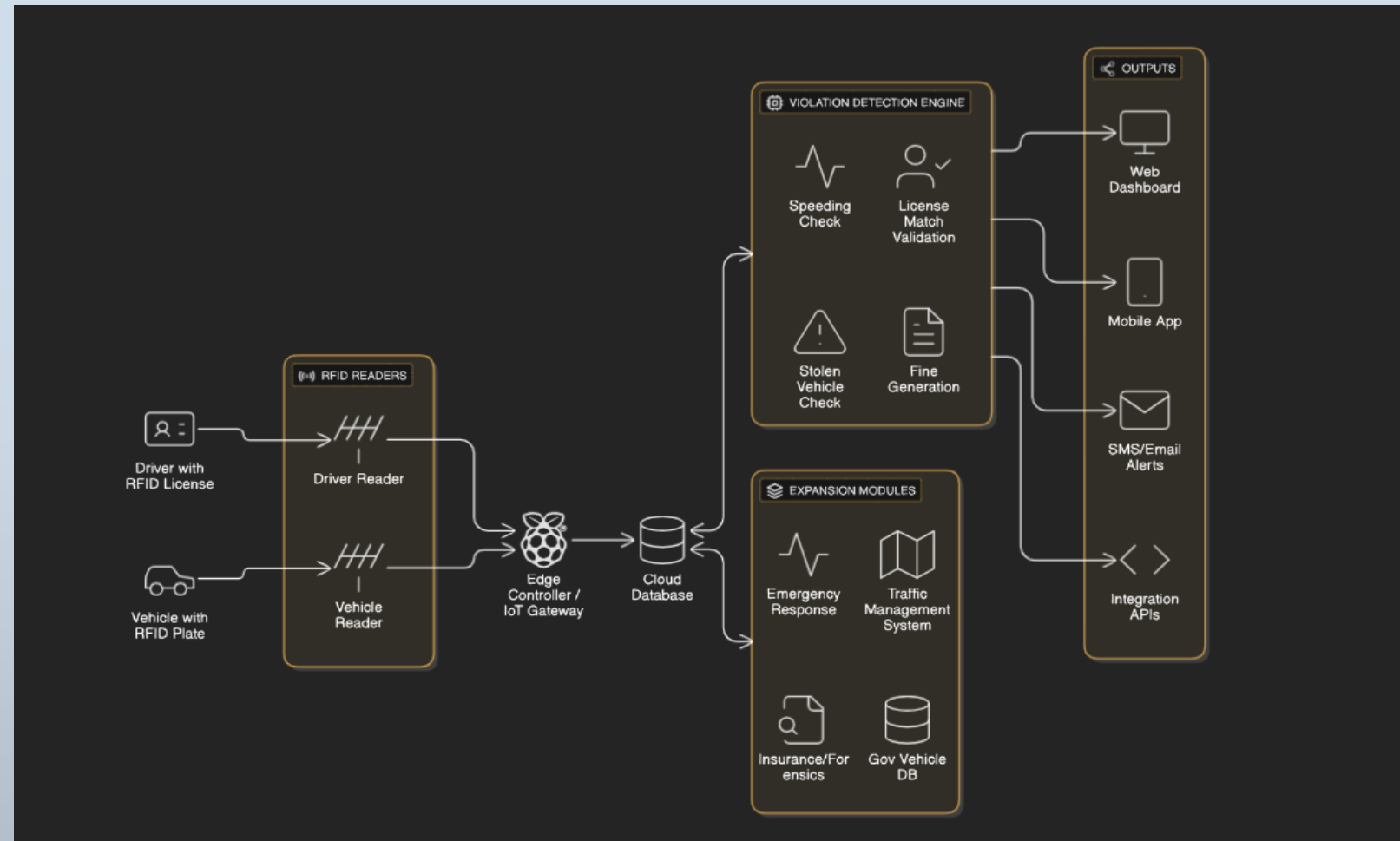
Evaluation Metrics

- Accuracy, Precision/Recall, F1-score, Confusion Matrix.
- Chart-based comparisons: predicted vs. actual violations.
- Target: >85% accuracy for prediction.

Data & Ethical Considerations

- Data from simulated traffic sets & RFID test collections.
- Only anonymized vehicle IDs used (no personal driver data).
- Compliance with privacy & data ethics standards.

System Overview



Dashboard & Driver Alert System



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Dashboard & Driver Alert System

Identified Challenges

- Current dashboards in traffic enforcement are reactive, not proactive.
- Lack of real-time visualization for violations.
- Difficulty in managing large datasets of violations securely.

Proposed Innovative Solution

- Develop a secure, admin-only web interface as a central control panel.
- Display real-time violation data with vehicle type, time, and driver details.
- Provide filtering & reporting (CSV export).
- Introduce multi-level Driver Alert System for instant, threshold-based feedback.

Limitations of Current Approaches

- Manual reporting → delays in enforcement.
- Limited filtering and reporting functions.
- No driver-level personalized feedback.

Dashboard & Driver Alert System

Core Knowledge Areas

- Web Development → Secure login, responsive dashboards.
- Database Management → Storing violation & driver records securely.
- Communication Systems → Real-time alerts via SMS.
- Data Integration → Linking image-processing results with dashboard displays.

Technologies Used

- Frontend: HTML5, CSS, React for responsive UI.
- Backend: APIs (Node.js) for data handling (integrated from team).
- Database: MySQL for violations, driver info & alerts.
- Alerting: SMTP for SMS notifications.
- Security: HTTPS, password hashing, admin-only authentication

Dashboard & Driver Alert System

Evaluation Metrics

- System Security: Only authorized admins can log in.
- Usability: Ease of dashboard navigation for non-technical officers.
- Alert Effectiveness: Timely delivery of SMS/Email alerts.
- Performance: Real-time updates with minimal latency.

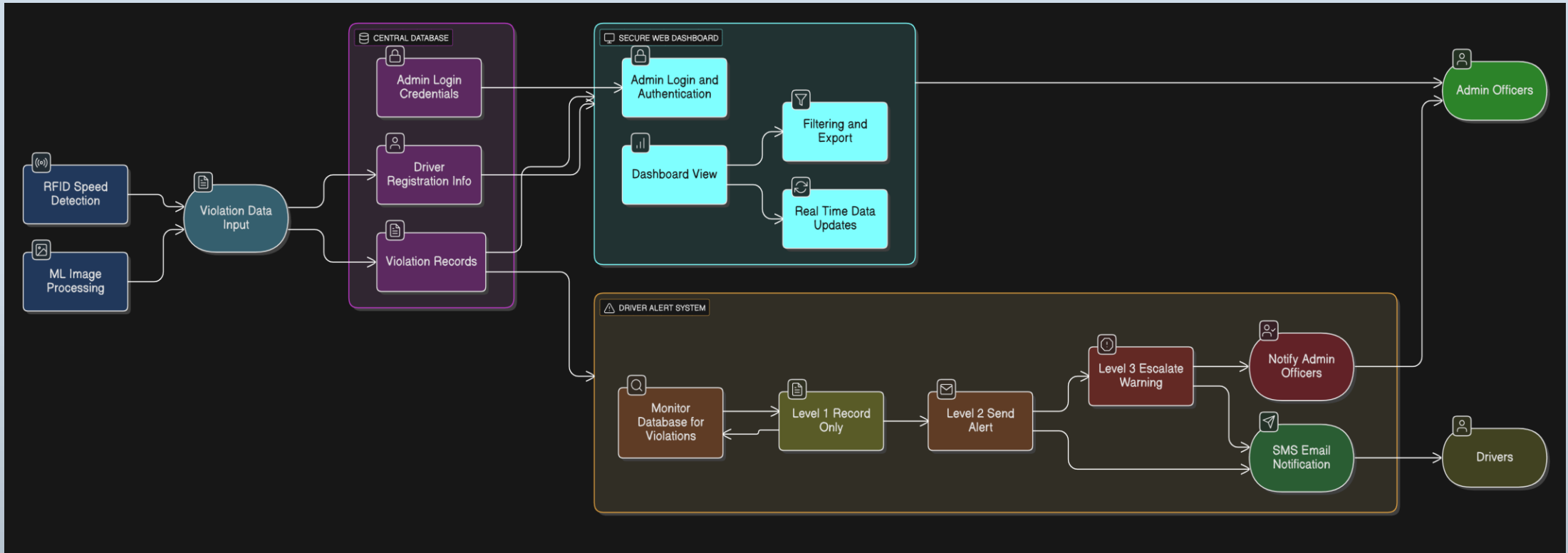
Data & Ethics

- Vehicle type data integrated from image-processing module.
- Driver details retrieved securely from RFID registration database.
- Only anonymized/necessary data is displayed; personal privacy protected.
- Alerts sent only after repeat violations → prevents unnecessary driver harassment.

Expected Outcomes

- Centralized, real-time monitoring dashboard for authorities.
- Clear visibility of violation details: vehicle type, time, driver contact.
- Reduction in repeat offences through multi-level driver alerts.
- Exportable official reports (CSV) for better decision-making.

System Overview



Smart IoT Solution for Vehicle Speed Monitoring via RFID License Tags



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Smart IoT Solution for Vehicle Speed Monitoring via RFID License Tags

Identified Challenges

- only catches speed at one location, not across a highway segment.
- Mid-segment speed checks, so driver/vehicle binding breaks in the middle.

Proposed Innovative Solution

- Use two RFID chips: one in the driver's license, one in the vehicle's plate.
- Checkpoints (entry, exit, midpoints) scan both tags - calculate speed

Limitations of Current Approaches

- Wind effects aren't detected → results ~99% accurate, not 100%.
- Tags/plates can be cloned or moved

Smart IoT Solution for Vehicle Speed Monitoring via RFID License Tags

Core Knowledge Areas

- Data Security & Privacy - Encrypted transmission, minimal personal data exposure
- IoT communication protocols– HTTP/MQTT for real-time data transfer.
- RFID Technology - UHF EPC Gen2 tag reading & dual-tag identification.

Technologies Used

- ESP32 Wi-Fi MCU – processing, timestamping, and network communication.
- UHF RFID Tags – card-type for driver license, on-metal/license plate tags for vehicles

Smart IoT Solution for Vehicle Speed Monitoring via RFID License Tags

Evaluation Metrics

- Number of vehicles correctly processed per minute/hour
- Tag Read Rate – percentage of successful reads at entry, exit, and midpoints.

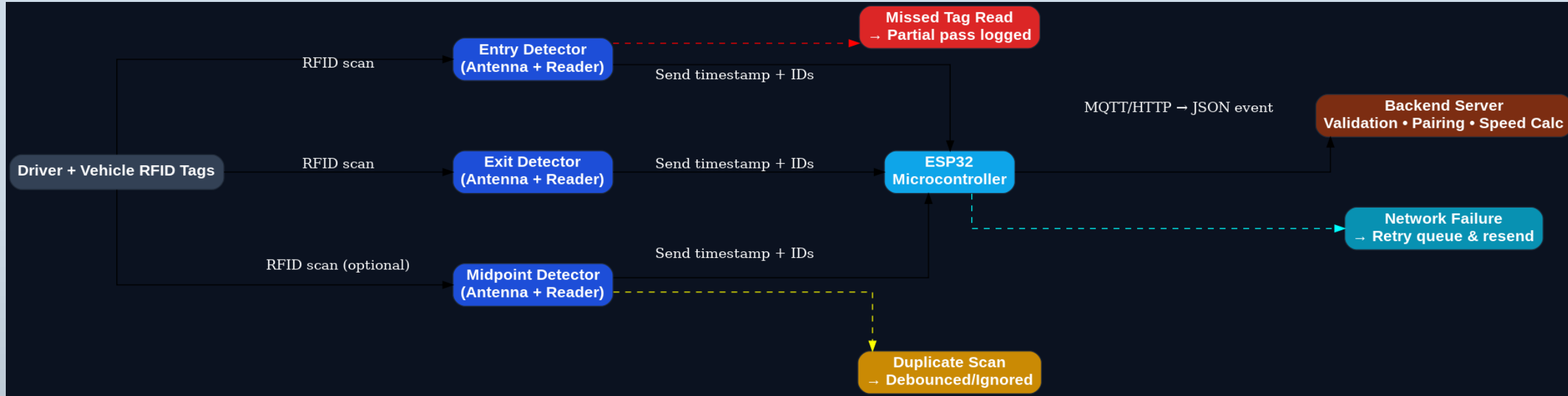
Data & Ethics

- Personal + vehicle details securely stored on RFID tags
- Encrypted communication (HTTPs/MQTT) to protect transmitted data.

Expected Outcomes

- Automated alerts for violations with minimal false positives.
- 99% speed accuracy across 100 m test segments.

Smart IoT Solution for Vehicle Speed Monitoring via RFID License Tags



Backend Service for Violation Detection and Real-Time Alert Dispatch with Vehicle Model Detection



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Backend Service for Violation Detection and Real-Time Alert Dispatch

Identified Challenges

- Most existing systems focus on storing violations but lack instant alerting mechanisms for enforcement.
- Most existing systems stop at brand-level recognition without going into exact model.

Proposed Innovative Solution

- Enable instant alert dispatch via message with no human involvement.
- Use image processing to recognize not only vehicle brand but also exact model (e.g., Honda Civic) enriching violation.

Limitations of Current Approaches

- Real-time alert system integrated with messaging for immediate enforcement.
- Image processing not just vehicle brands, but also exact model.

Backend Service for Violation Detection and Real-Time Alert Dispatch

Core Knowledge Areas

- Designing and implementing RESTful APIs to collect and manage RFID-based speed data.
- Applying zone-based speed violation logic for accurate detection.
- Developing and integrating alert mechanisms for timely enforcement.
- Using DTK software to identify vehicle model.

Technologies Used

- Real-Time Messaging Systems – For instant alert notifications
- DTK Software- To vehicle type recognition through image processing.

Backend Service for Violation Detection and Real-Time Alert Dispatch

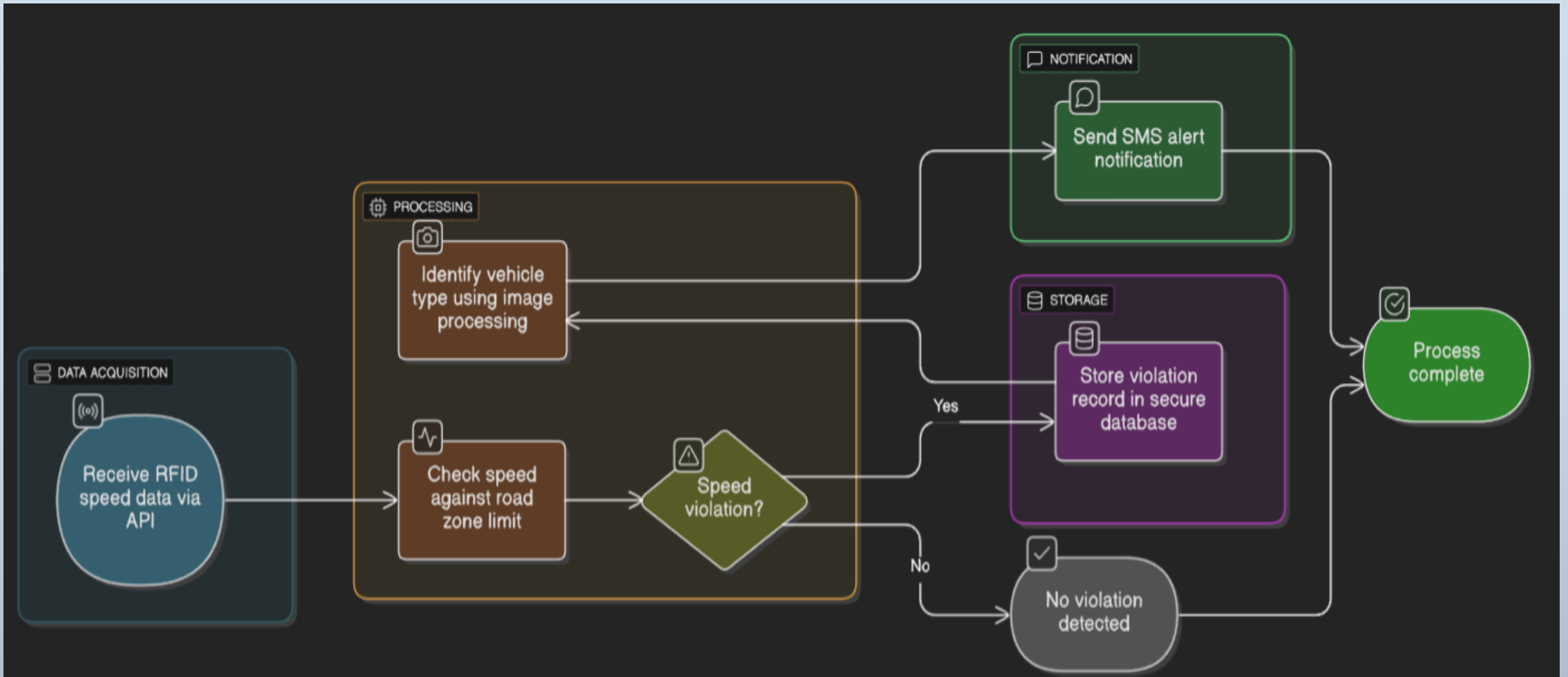
Evaluation Metrics

- Every speed-violating vehicle triggers an instant message alert containing the correct vehicle details with brand and exact model

Data & Ethical Considerations

- Violation records and images are stored in encrypted databases with restricted access.

Backend Service for Violation Detection and Real-Time Alert Dispatch



IoT & ML-based Vehicle Speed Monitoring with RFID-Enabled License Validation

Investment & Cost Recovery

- ✓ Initial Investment: Rs. 2M (RFID devices, servers, development)
- ✓ Operating Cost: Rs. 250,000/month
- ✓ Revenue Model:
 - Govt. contracts (Rs. 100,000/junction)
 - Annual subscription/licensing fees
 - Data analytics services & insurance partnerships

Target Market

- ✓ Traffic Police & Road Safety Authorities
- ✓ Municipal Councils (25+ cities, Smart City projects)
- ✓ Highway Authorities
- ✓ Insurance & Transport Companies

Thank you